

Hegemony of the US dollar for the Global Economy

Shiping Liu

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Background

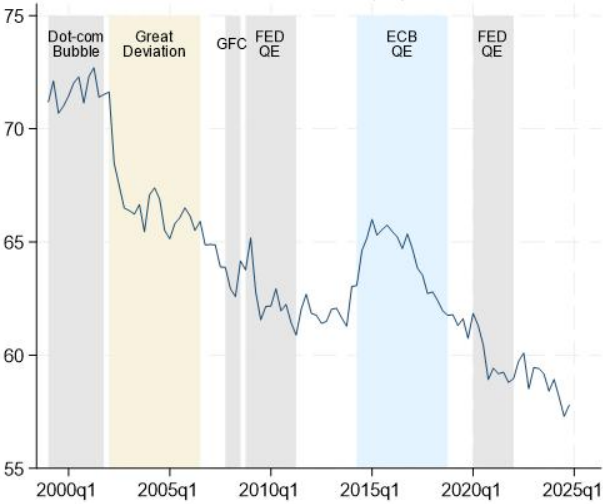
- Research Question
 - To what extent do the currency shares in COFER respond to changes in the exchange rates?
 - If they do, is the sign of the response consistent with the asset-pricing model?
 - How do the results change across different sample periods?
- Theoretical Frameworks
 - Valuation Effect (VE): If a currency **appreciates against USD**, share of this currency in COFER **↑**.
 - Substitutability: Currencies are viewed as **rivals** or **competitors** in a portfolio (consistent with VE).
 - Complementarity: Currencies are viewed as **allies** or parts of a single bundle (inconsistent with VE).
- Potential Application
 - Uncover substitutability and complementarity among currencies.
 - Guide monetary policies for central banks.
 - Provide a method to signal economic recessions.

Preview of Conclusions & Takeaways

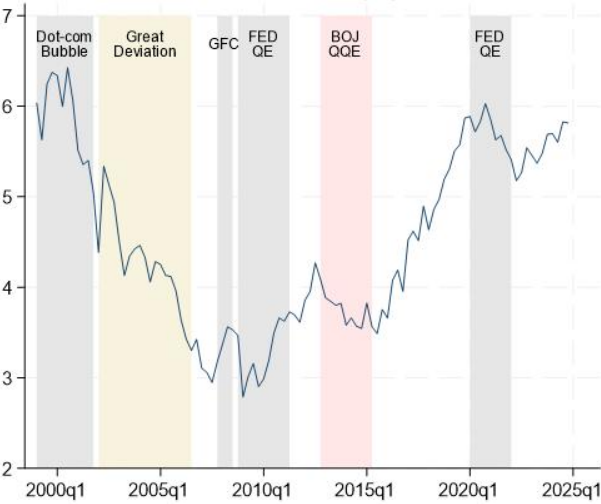
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 - To what extent do the currency shares in COFER respond to changes in the exchange rates?
 - If they do, is the sign of the response consistent with the asset-pricing model?
 - How do the results change across different sample periods?
- Conclusions
 - COFER currency shares respond significantly to exchange rate fluctuations.
 - Generally, the Euro is the sole **strong substitute** to the USD, consistent with the asset-pricing model.
 - During **2008 Financial Crisis**, all other currencies of interest (Yen, Euro, Pound) become substitutes of USD, i.e., aligning with valuation effect and asset-pricing model.
 - The elasticities of currency shares are not constant, which change due to **monetary policies** (e.g., QE, QQE) and **economic cycles**.
- Takeaways
 - A **higher sensitivity** of currency share to exchange rates can serve as a signal of recessions.

Data Trends

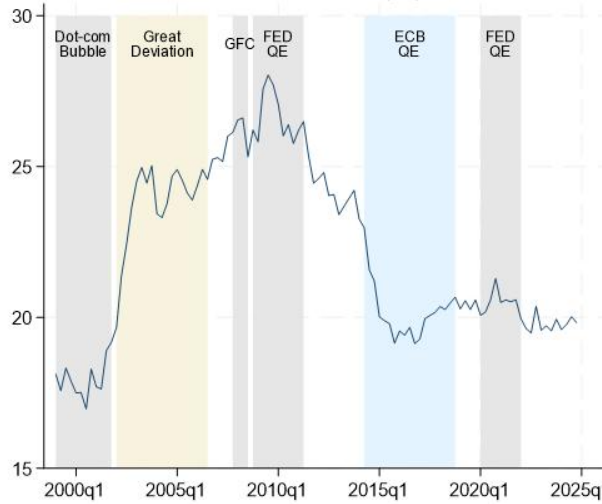
USD Share (%)



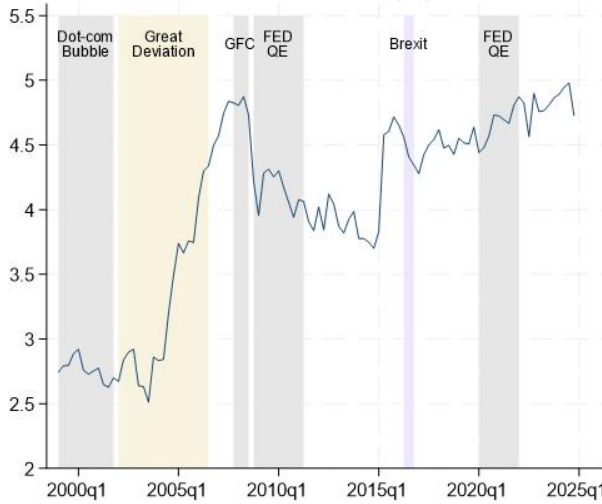
Yen Share (%)



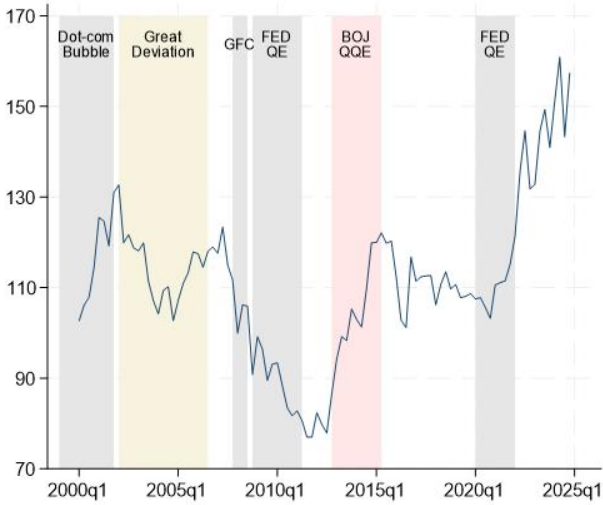
Euros Share (%)



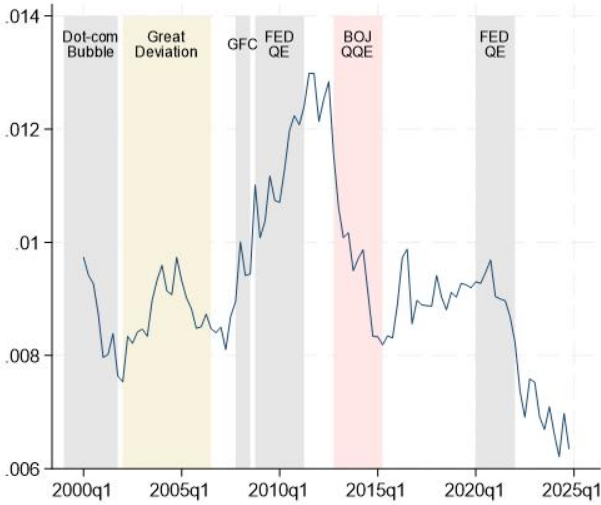
Pounds Share (%)



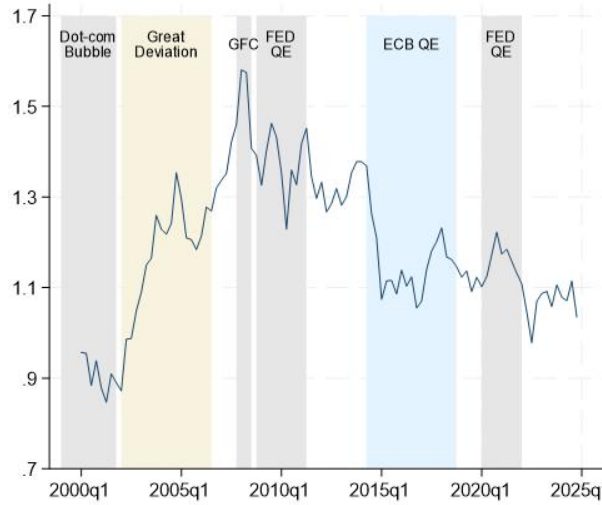
Yen/USD



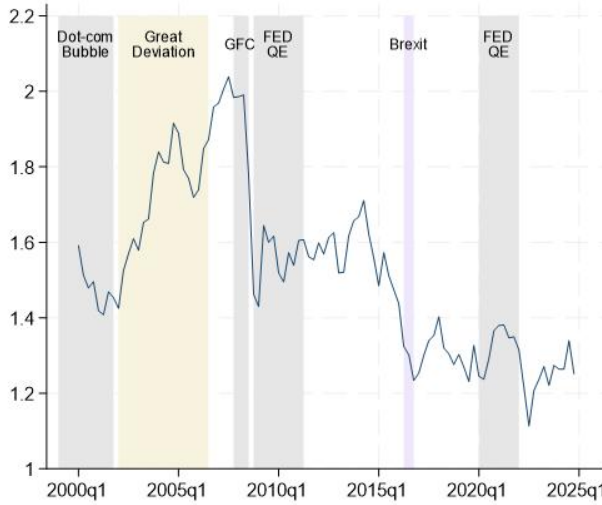
USD/Yen



USD/Euro



USD/Pound



Methodology

- Data
 - IMF COFER dataset from 2000Q1 to 2024Q4
- Linear-Linear Model
 - $Share\ of\ \$_t = \alpha + \beta \cdot Share\ of\ \$_{t-1} + \gamma_1 \cdot E_{Yen/\$, t} + \gamma_2 \cdot E_{\$/Euro, t} + \gamma_3 \cdot E_{\$/Pound, t} + \varepsilon_t$
 - $\hat{\gamma}_{1, upper} = \hat{\gamma}_1 + 1.96 \cdot SE_{\hat{\gamma}_1}, \hat{\gamma}_{1, lower} = \hat{\gamma}_1 - 1.96 \cdot SE_{\hat{\gamma}_1}$
 - $Elasticity\ to\ E_{Yen/\$}_t = \hat{\gamma}_1 \cdot \frac{E_{Yen/\$, t}}{Share\ of\ \$_t}$
 - $95\%\ CI = [\hat{\gamma}_{1, upper} \cdot \frac{E_{Yen/\$, t}}{Share\ of\ \$_t}, \hat{\gamma}_{1, lower} \cdot \frac{E_{Yen/\$, t}}{Share\ of\ \$_t}]$
- Log-Log Model (Constant Elasticity Model)
 - $\ln(Share\ of\ \$_t) = \alpha + \beta \cdot \ln(Share\ of\ \$_{t-1}) + \gamma_1 \cdot \ln(E_{Yen/\$, t}) + \gamma_2 \cdot \ln(E_{\$/Euro, t}) + \gamma_3 \cdot \ln(E_{\$/Pound, t}) + \varepsilon_t$
 - $Elasticity\ to\ E_{Yen/\$}_t \equiv \hat{\gamma}_1$

Main Findings

- Currency Share Inertia
- Consistency to Valuation Effect
 - Yen depreciates → USD Share ↓ (X)
 - Euro appreciates → USD Share ↓ (✓)
 - Pound appreciates → USD Share ↑ (X)
- Currency Relationship

USD & Yen	Weak Complements (-0.016)
USD & Euro	Strong Substitutes (-4.429)
USD & Pound	Complements (2.248)
Yen & Pound	Substitutes (-0.361)
Euro & Pound	Complements (0.786)

Table 1: Regression Results of Currency Shares on Exchange Rates

	Share (%)				Log of Share			
	(1) USD (\$)	(2) Yen	(3) Euro	(4) Pound	(5) USD (\$)	(6) Yen	(7) Euro	(8) Pound
L. %\$	0.848*** (19.20)							
Yen/\$	-0.016** (-2.26)	-0.001 (-0.66)	0.001 (0.31)	0.002 (1.53)				
\$/Euro	-4.429*** (-3.17)	0.084 (0.32)	1.136 (1.42)	0.081 (0.32)				
\$/Pound	2.248*** (2.82)	-0.361** (-2.51)	0.786** (2.06)	0.178 (1.30)				
L. %Yen		0.928*** (23.49)						
L. %Euros			0.877*** (18.53)					
L. %Pounds				0.971*** (25.04)				
L. Log %\$					0.844*** (18.56)			
Log Yen/\$					-0.024** (-2.09)	-0.048 (-1.05)	0.006 (0.28)	0.081** (2.09)
Log \$/Euro					-0.079*** (-3.20)	-0.006 (-0.09)	0.050 (1.19)	0.078 (1.02)
Log \$/Pound					0.055*** (2.90)	-0.126** (-2.43)	0.056** (2.11)	0.042 (0.75)
L. Log %Yen						0.921*** (22.04)		
L. Log %Euros							0.882*** (18.02)	
L. Log %Pounds								0.954*** (24.82)
Constant	13.173*** (3.40)	0.896* (1.67)	0.046 (0.05)	-0.475* (-1.89)	0.749*** (3.44)	0.395* (1.75)	0.304 (1.55)	-0.344** (-2.11)
R ²	0.954	0.940	0.955	0.953	0.953	0.936	0.955	0.958
adj. R ²	0.952	0.938	0.953	0.951	0.951	0.933	0.953	0.956
N	100	100	100	100	100	100	100	100

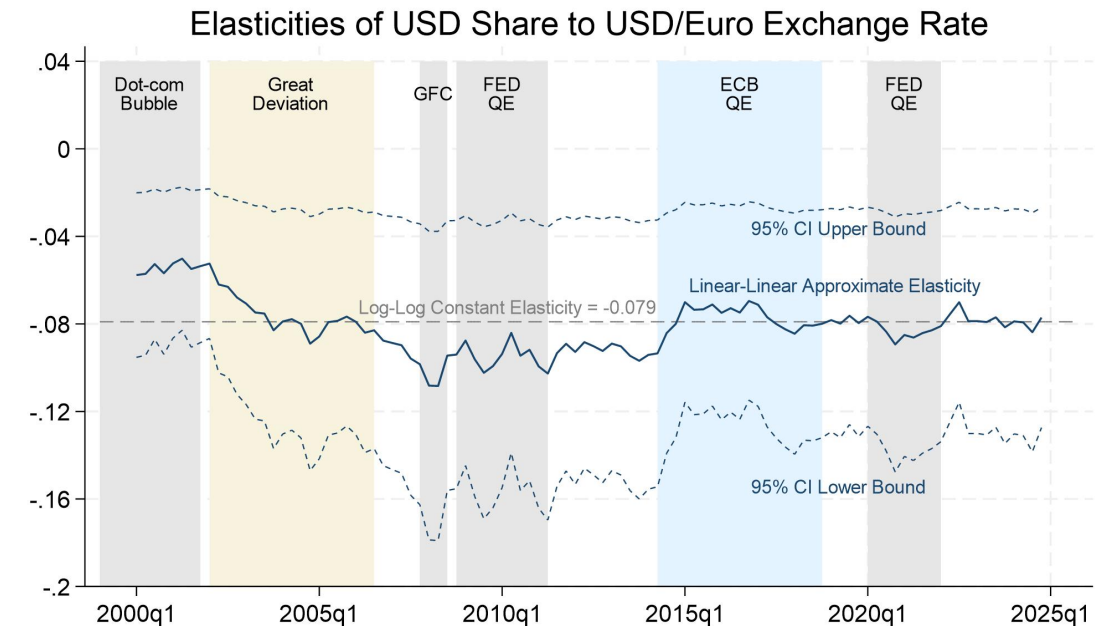
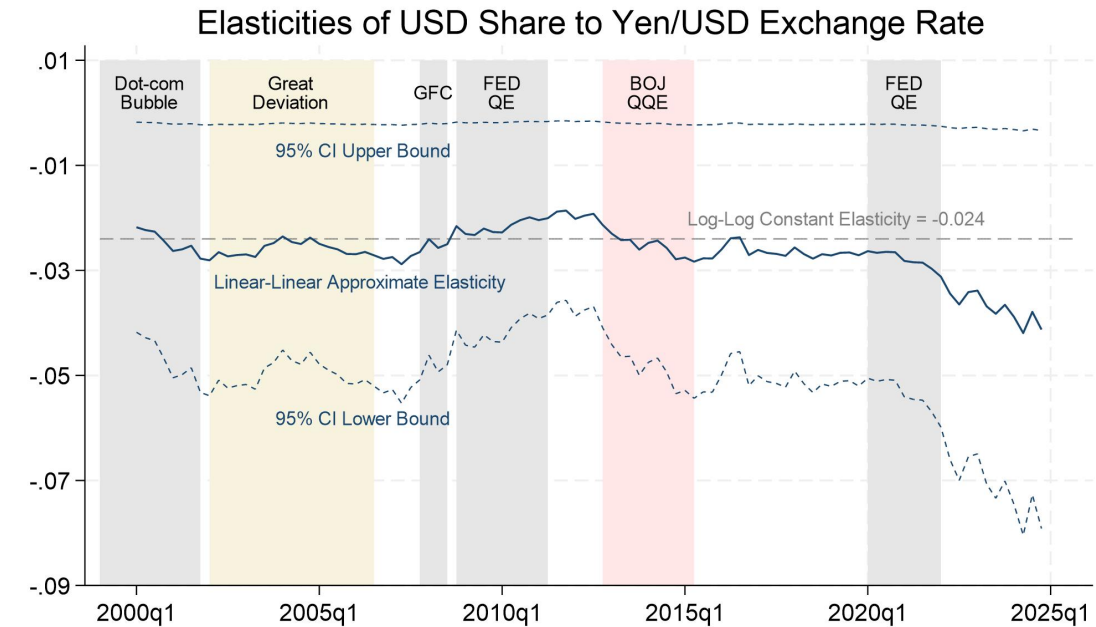
t statistics in parentheses

Note: Columns 1-4 are standard linear models; Columns 5-8 are log-log models. Lagged dependent variables included.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Main Findings

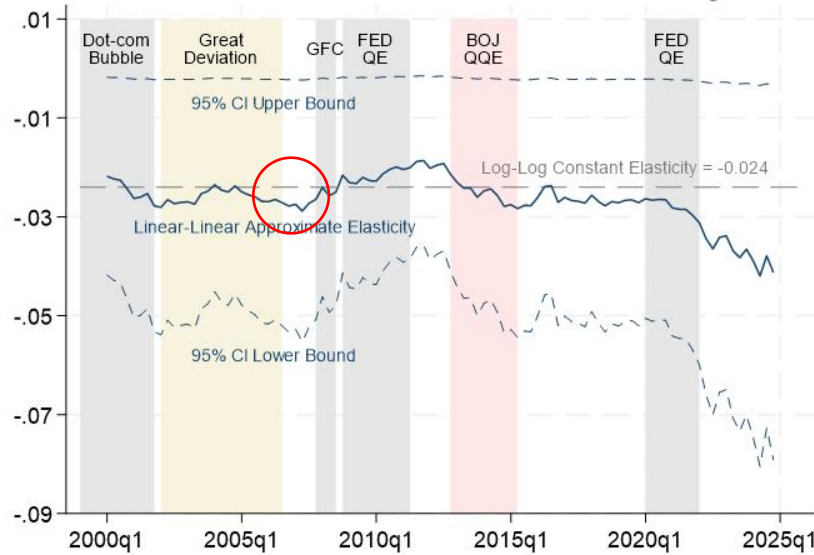
- Common Characteristics
 - Absolute value of elasticity $\uparrow \rightarrow$ USD share sensitive $\uparrow \rightarrow$ **stronger** relationship
 - The Great Deviation period makes USD & Yen stronger complements and USD & Euro stronger substitutes.
 - GFC and the following FED QE make USD **more substitutable** between both Yen and Euro (See Page 9).
- USD & Yen (Complements)
 - BOJ QQE makes USD & Yen stronger complements.
 - After BOJ QQE, the complementarity between USD & Yen **keeps going stronger**.
- USD & Euro (Substitutes)
 - ECB QE makes USD & Euro weaker substitutes.



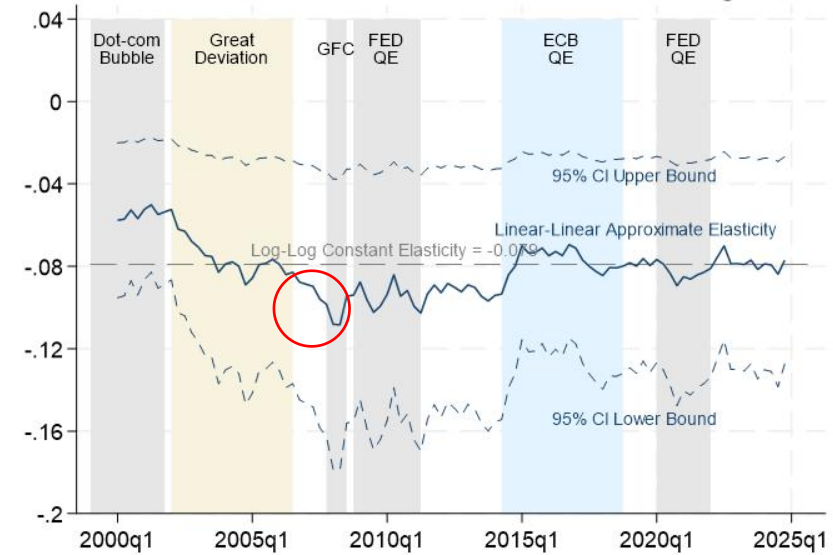
Main Findings

- The elasticities of **every** pair of currency share and exchange rate experience a **greater absolute value** before the arrival of 2008 Financial Crisis.
- A more sensitive elasticity is a harbinger of a great recession.

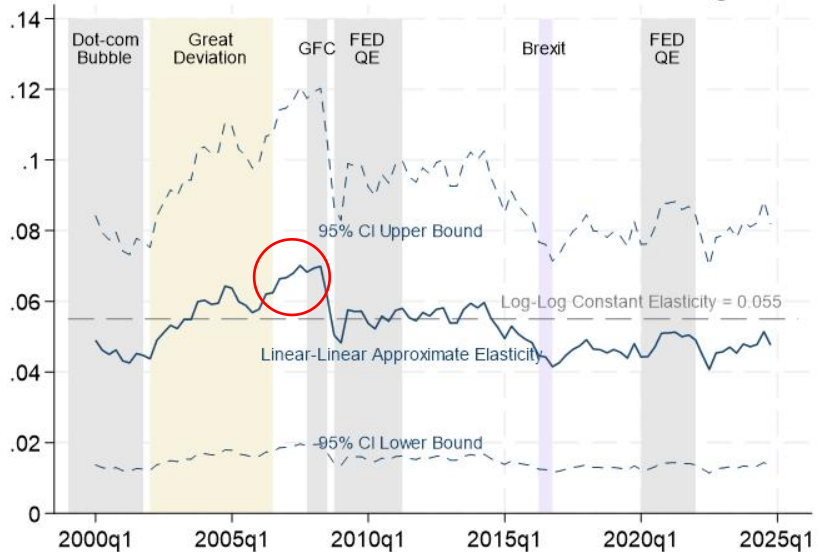
Elasticities of USD Share to Yen/USD Exchange Rate



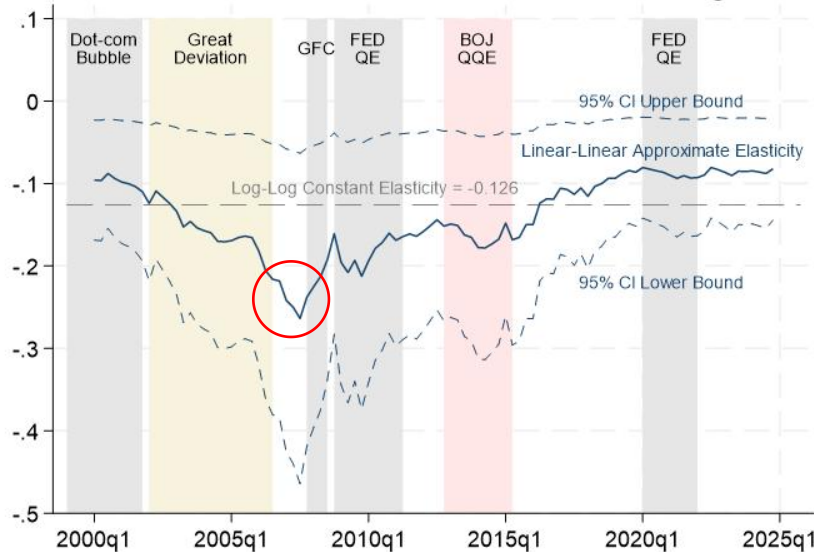
Elasticities of USD Share to USD/Euro Exchange Rate



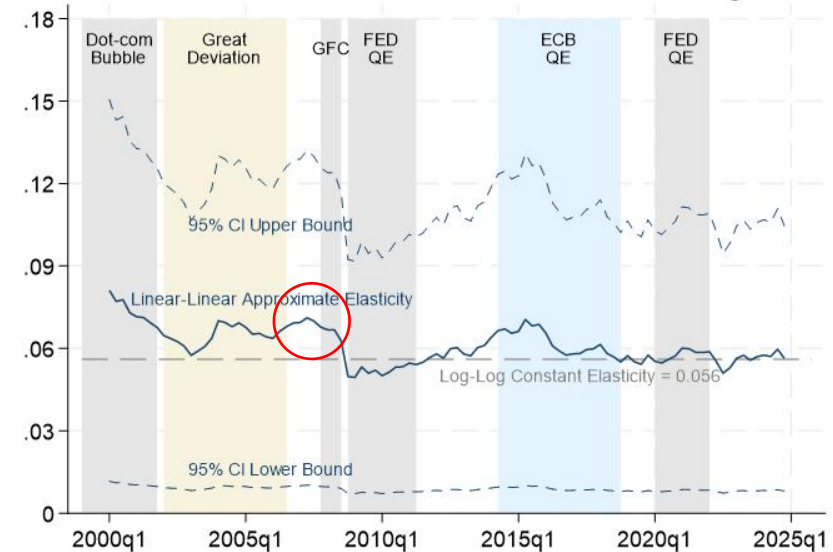
Elasticities of USD Share to USD/Pound Exchange Rate



Elasticities of Yen Share to USD/Pound Exchange Rate



Elasticities of Euro Share to USD/Pound Exchange Rate



Sensitivity Analysis

- Rolling Window Regressions

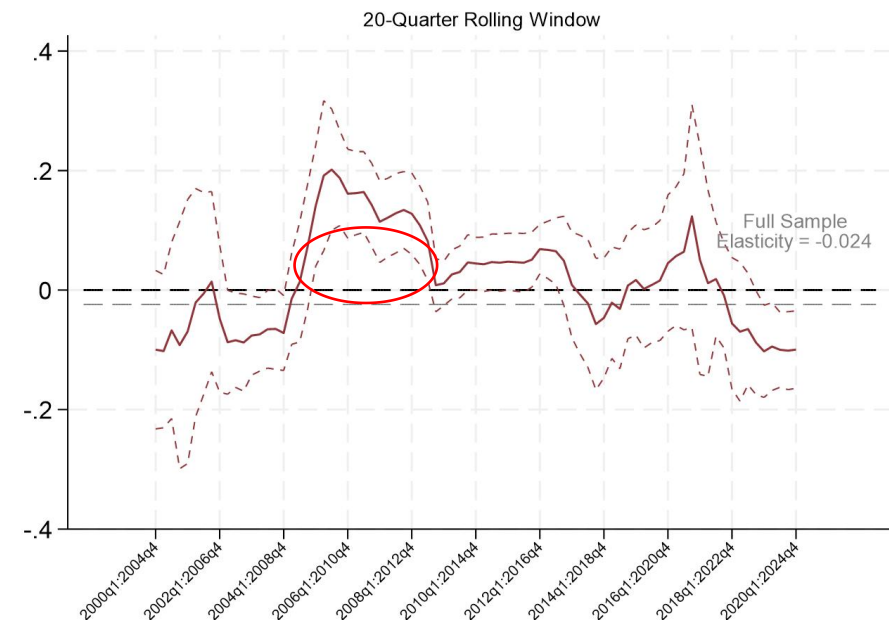
- From 2000Q1, choose the following 20 or 40 quarters as a rolling window, until 2024Q4.
- Log-Log Model (Constant Elasticity Model):
 - $\ln(\text{Share of } \$_t) = \alpha + \beta \cdot \ln(\text{Share of } \$_{t-1}) + \gamma_1 \cdot \ln(E_{Yen/\$, t}) + \gamma_2 \cdot \ln(E_{\$/Euro, t}) + \gamma_3 \cdot \ln(E_{\$/Pound, t}) + \varepsilon_t$
- Elasticity to $E_{Yen/\$}_t \equiv \hat{\gamma}_1$ as a function of sample periods.

- Results

- The 20-quarter rolling window regressions have generally wider 95% CI bands than 40-quarter rolling window regressions have.
- Even if the estimated elasticity of the regression over full sample is significant -0.024 (seen in Table 1), it goes significantly positive, especially in those subsamples including the GFC. This confirms the analyses in Page 7: GFC and the following FED QE make USD **more substitutable** between Yen.
- This phenomenon also appears in USD & Euro, USD & Pound (See Page 10).

Constant Elasticities over Rolling Window Regressions

USD Share vs Yen/USD (Log-Log Model)

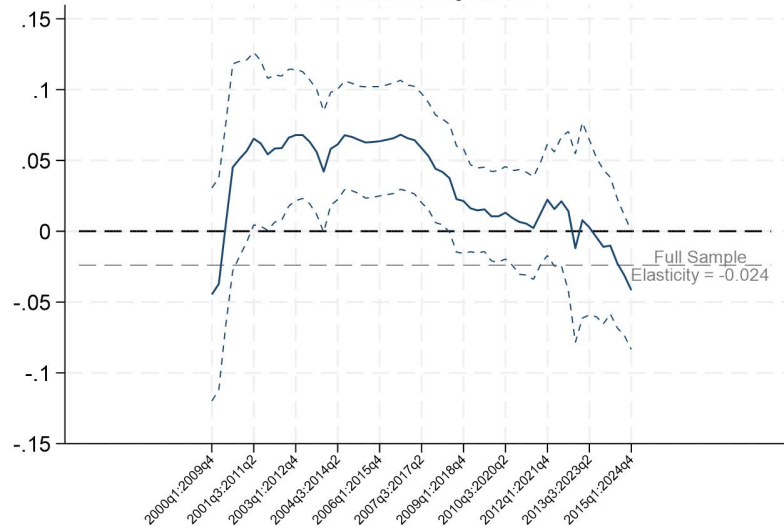


Sensitivity Analysis

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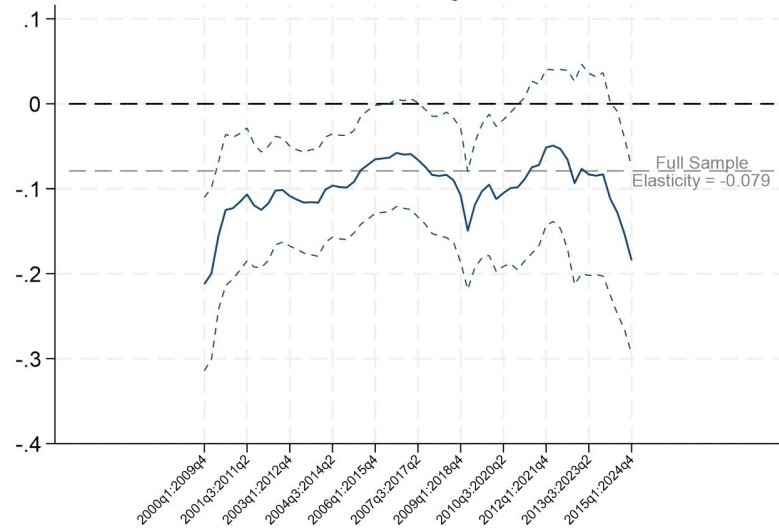
40-Quarter Rolling Window



Constant Elasticities over Rolling Window Regressions

USD Share to USD/Euro Exchange Rate (Log-Log Model)

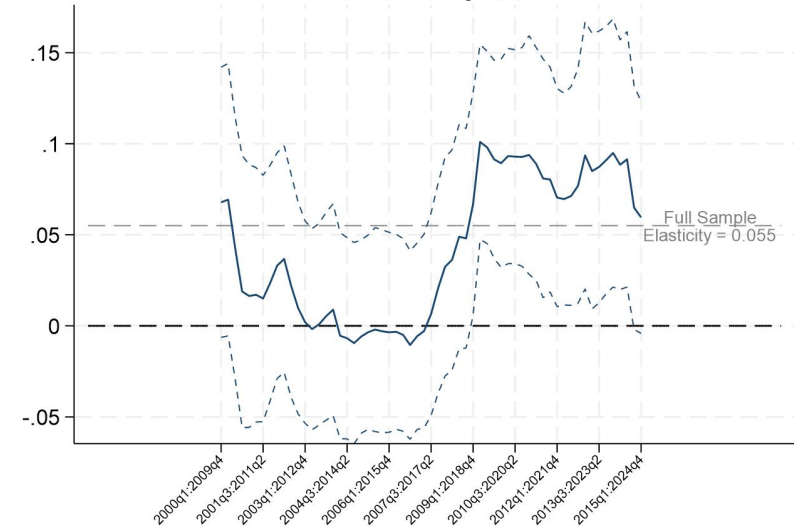
40-Quarter Rolling Window



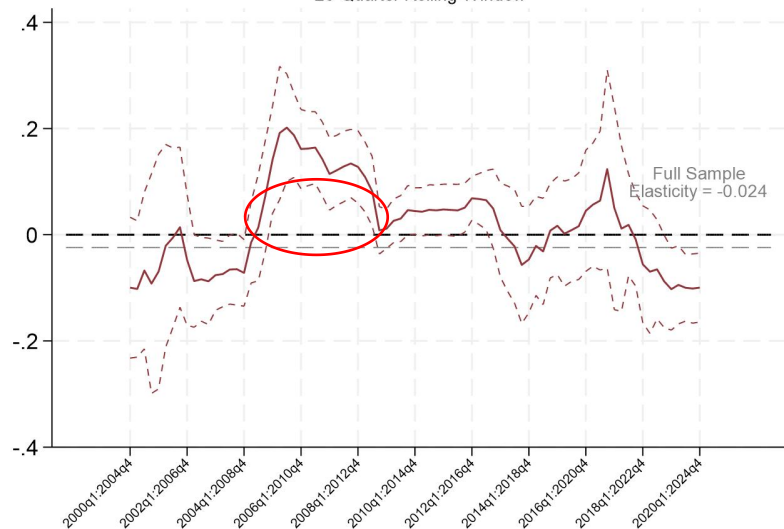
Constant Elasticities over Rolling Window Regressions

USD Share vs USD/Pound (Log-Log Model)

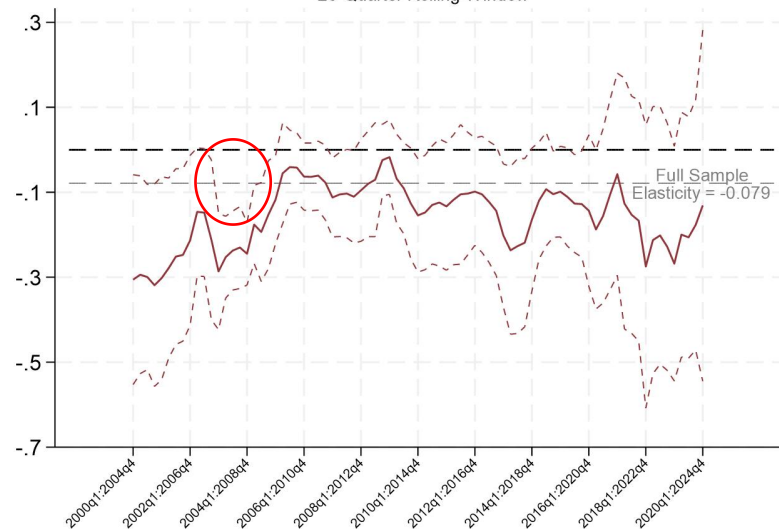
40-Quarter Rolling Window



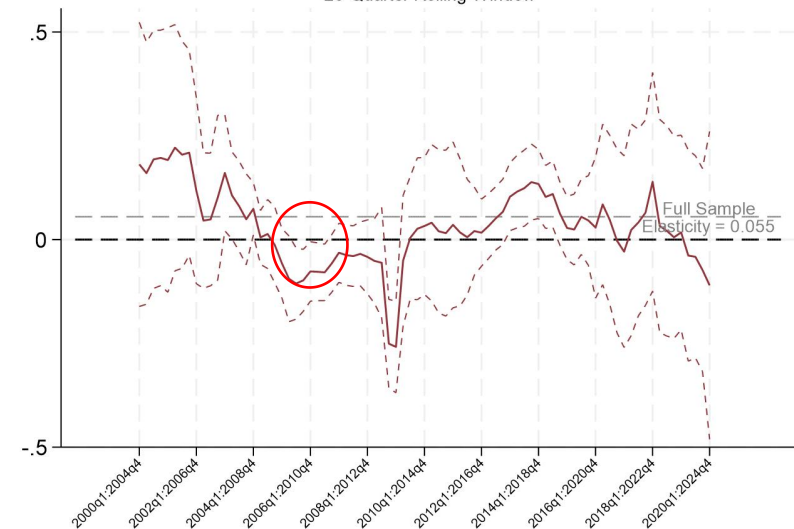
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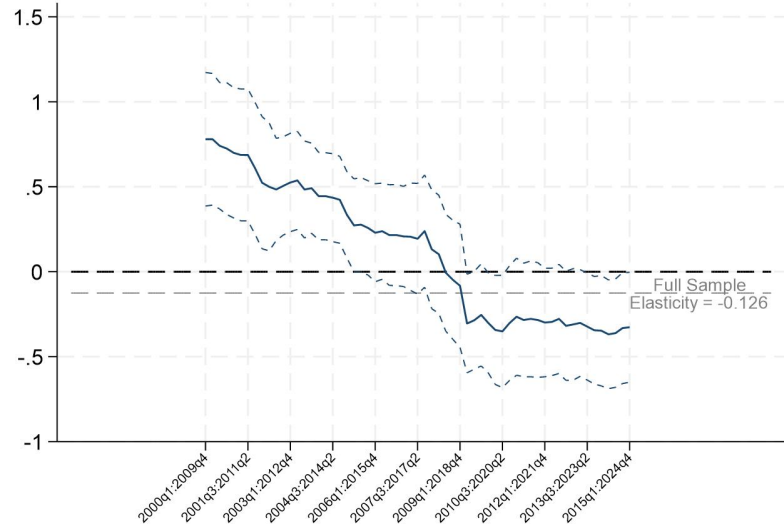


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Yen Share vs USD/Pound (Log-Log Model)

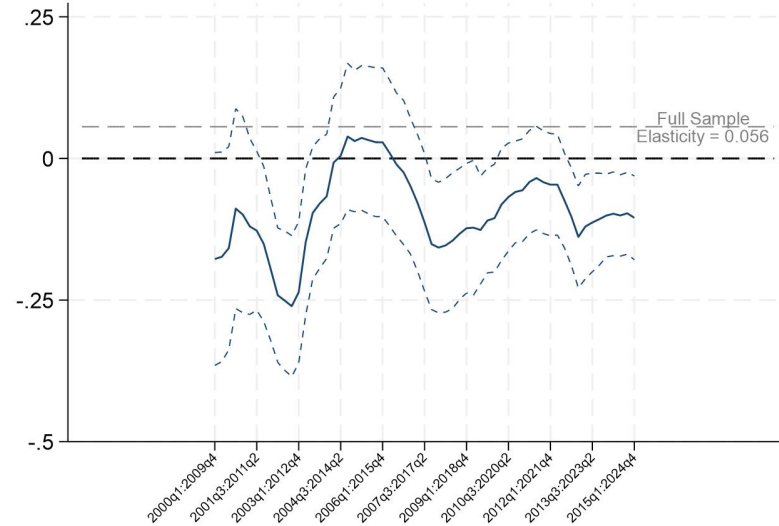
40-Quarter Rolling Window



Constant Elasticities over Rolling Window Regressions

Euro Share vs USD/Pound (Log-Log Model)

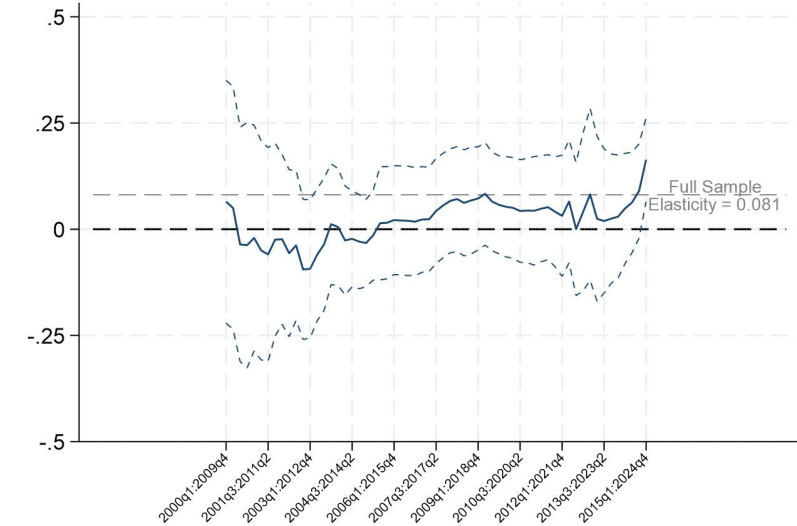
40-Quarter Rolling Window



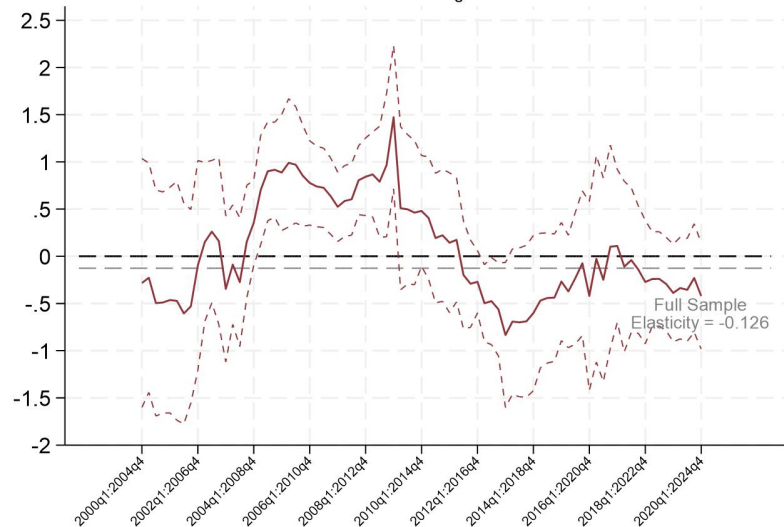
Constant Elasticities over Rolling Window Regressions

Pound Share vs Yen/USD (Log-Log Model)

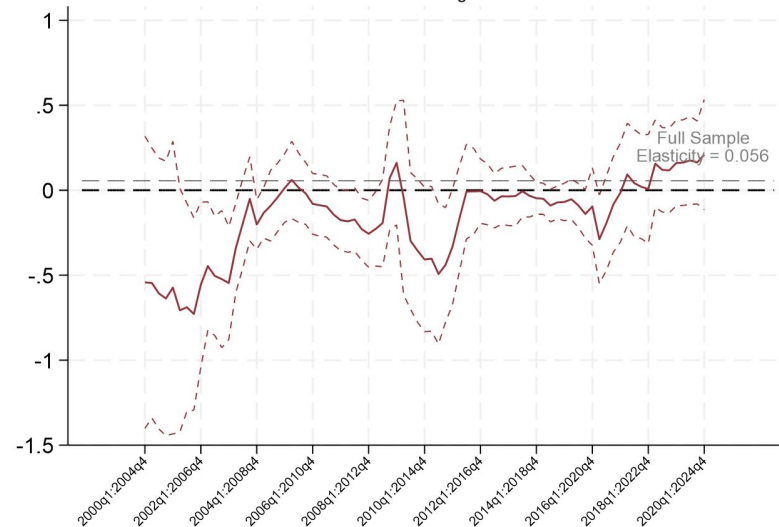
40-Quarter Rolling Window



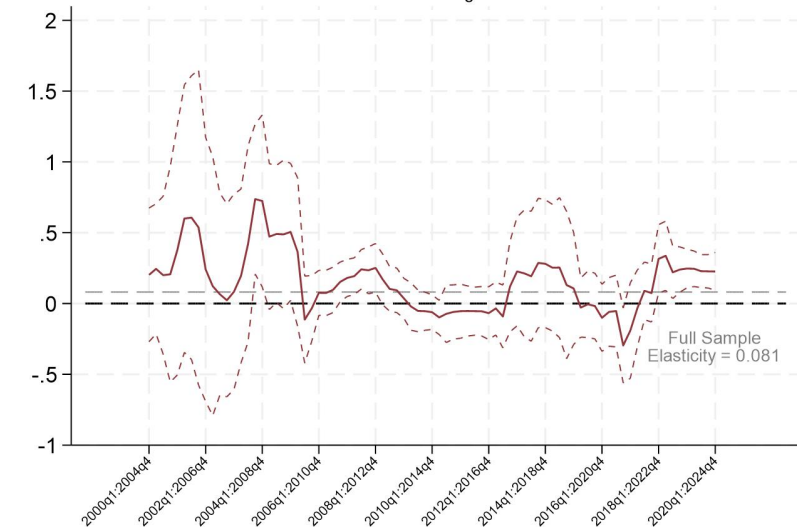
20-Quarter Rolling Window



20-Quarter Rolling Window



20-Quarter Rolling Window



Conclusions & Takeaways

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